Analyzing the Relationship Between COVID-19 Vaccination Coverage and Mortality Through Time-Lagged and Cross-Country Analysis

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***Abstract*—COVID-19 vaccination programmes generated substantial public health data that can be examined alongside mortality records to understand how the two indicators change over time. This study investigates the observed relationship between vaccination coverage and mortality across geographical locations using publicly available COVID-19 data.**

Two datasets containing vaccination and mortality information from January 2020 to April 2023 were combined using geographical identifiers and observation dates. The resulting merged dataset contained 302,749 records. After excluding aggregate geographical entries, 288,347 records covering 243 geographical locations were retained. Fully vaccinated population percentage was selected as the primary vaccination measure, while seven-day smoothed new deaths per million was used as the mortality measure.

The analysis includes country-level time-series visualization, Pearson and Spearman correlation, time-lagged comparisons at 14, 28 and 42 days, and weekly-change analysis across geographical locations. Multiple statistical comparisons were adjusted using the Benjamini–Hochberg procedure. An interactive dashboard was developed to support country selection, date filtering, lag comparison and visualization.

For India, negative correlations were observed at all three investigated vaccination lead times. However, the cross-location weekly-change analysis showed median Pearson and Spearman correlations close to zero. Only three negative Pearson associations remained significant after false discovery rate adjustment, while none of the negative Spearman associations passed the adjustment threshold.

The findings demonstrate that the observed relationship varies according to the analytical approach and geographical location. The results are exploratory and do not establish that vaccination alone caused changes in mortality.

***Index Terms*—COVID-19, vaccination coverage, mortality analysis, time-series analysis, Pearson correlation, Spearman correlation, lagged analysis, cross-country comparison, public health data, interactive dashboard.**

I. Introduction

The COVID-19 pandemic created a major challenge for healthcare systems and public health authorities across the world. During the pandemic, information about infections, vaccination programmes and reported deaths was collected from different geographical locations and made available through public datasets.

These datasets provide an opportunity to examine how pandemic indicators changed during different periods. However, the availability of large amounts of data does not automatically make the relationships between different indicators easy to understand.

Vaccination coverage and mortality are two important indicators of pandemic progression. Vaccination coverage describes the reported proportion of a population that has completed the initial vaccination protocol, whereas mortality provides information about reported deaths. Their relationship can be examined across time and geographical locations to identify patterns within the available data.

Several studies have examined the effectiveness of COVID-19 vaccines through individual-level and population-level observational studies. Research conducted in Israel and other settings reported reductions in severe disease and COVID-19-related outcomes among vaccinated populations [7]–[9].

Global modelling studies have also investigated the wider impact of vaccination programmes on pandemic mortality [5].

These studies use different research designs from a country-level correlation analysis. In particular, a correlation between vaccination coverage and mortality does not directly measure individual vaccine effectiveness. It also does not separate vaccination from other factors that may influence mortality.

A further consideration is the time between vaccination and a subsequently recorded mortality observation. Comparing both indicators on the same date may not capture all possible temporal patterns. Examining different time intervals provides another perspective on their observed association.

This study therefore focuses on examining the relationship between COVID-19 vaccination coverage and mortality through time-lagged and cross-country analysis.

The central research question is:

> What relationship can be observed between COVID-19 vaccination coverage and mortality trends across different geographical locations and time periods?

The main objectives are:

- To combine public vaccination and mortality datasets using geographical identifiers and dates.
- To examine vaccination coverage and mortality trends over a common timeline.
- To calculate Pearson and Spearman correlation coefficients for selected geographical locations.
- To investigate relationships using vaccination lead intervals of 14, 28 and 42 days.
- To compare weekly changes in vaccination coverage with subsequent changes in mortality across geographical locations.
- To develop an interactive dashboard that allows users to explore the observed patterns through country, date and lag selection.

The main contribution is an integrated analytical workflow that combines data preparation, time-lagged statistical analysis, cross-location comparison and interactive visualization.

II. Related Work

COVID-19 data visualization and vaccination effectiveness have been investigated through different research approaches.

Dong et al. introduced an interactive web-based dashboard for monitoring COVID-19 cases in real time [1]. Their work demonstrated how information collected from multiple sources could be presented through a centralized geographical visualization platform.

Ivanković et al. evaluated 158 public COVID-19 dashboards from 53 countries [2]. Their assessment examined the features available in existing systems and highlighted the importance of useful indicators, clear presentation and analytical interaction.

Mathieu et al. developed a global database of COVID-19 vaccination information, providing standardized data for examining vaccination progression across geographical locations [3]. The corresponding Our World in Data repository also provides historical COVID-19 data containing vaccination, mortality and population-adjusted indicators [4].

Research on vaccine effectiveness provides additional context for studying vaccination and mortality. Watson et al. used mathematical modelling to estimate the global impact of the first year of COVID-19 vaccination programmes [5]. Feikin et al. conducted a systematic review and meta-regression to investigate how vaccine effectiveness changed with time after vaccination [6].

Haas et al. studied the effectiveness of the BNT162b2 vaccine using national surveillance data from Israel and examined outcomes including hospitalization and deaths [7]. Dagan et al. investigated the effectiveness of the BNT162b2 vaccine in a nationwide mass vaccination setting [8]. Tenforde et al. examined associations between mRNA vaccination, hospitalization and disease severity [9].

These studies provide evidence from different research designs. Their findings cannot be directly transferred to an ecological analysis of aggregated country-level data because the measurement units, populations and statistical methods differ.

Msemburi et al. also highlighted the limitations of reported COVID-19 mortality by examining excess mortality associated with the pandemic [10]. Differences in death registration, testing capacity and reporting practices are important when comparing mortality observations across geographical locations.

The present study uses publicly available vaccination and mortality series to investigate their observed association through multiple time-lagged comparisons and weekly-change analysis.

III. Research Gap and Problem Formulation

Existing COVID-19 data resources provide extensive information about vaccination coverage and mortality. However, the interpretation of these indicators requires more than displaying their values independently.

This study focuses on four analytical considerations.

A. Separate Examination of Indicators

Vaccination and mortality are often presented as separate statistical indicators. Examining both measures together allows their historical patterns to be compared within the same geographical and temporal scope.

B. Temporal Relationship

A relationship may not be visible when vaccination and mortality are compared only on the same date. Different vaccination lead intervals can be examined to investigate whether earlier coverage is associated with subsequent mortality.

C. Geographical Variation

A pattern observed in one location may not be consistent across other locations. Differences in reporting, population structure, vaccination progression and epidemic conditions make cross-location comparison important.

D. Level-Based Versus Change-Based Analysis

Vaccination coverage generally accumulates over time, whereas mortality can increase and decrease during successive pandemic waves. Strong correlation between the levels of two variables may therefore partly reflect their shared time trends.

Examining changes in vaccination coverage and mortality provides a complementary analytical approach.

Based on these considerations, the study compares the historical levels of the two indicators and their subsequent weekly changes.

IV. Dataset Description

The analysis uses two publicly available CSV datasets containing country-level and geographical time-series information.

The vaccination dataset contains observations related to vaccine administration and population coverage. The mortality dataset contains reported deaths, daily death counts and population-adjusted mortality measures.

The dataset structure and vaccination indicators are consistent with the documented Our World in Data COVID-19 data resources [3], [4].

A. Dataset Characteristics

Both datasets cover the period from January 1, 2020, to April 18, 2023.

Table I presents the basic characteristics of the input files.

TABLE I
Characteristics of the Primary Datasets

Property	Vaccination	Mortality
Rows	302,749	302,749
Columns	45	26
Start date	2020-01-01	2020-01-01
End date	2023-04-18	2023-04-18
Unique locations	255	255
Duplicate keys	0	0

The datasets were joined using three common fields: ISO geographical code, location name and date.

$$K = \{\text{iso_code}, \text{location}, \text{date}\} \tag{1}$$

An inner join produced 302,749 merged observations and 68 columns.

After removing aggregate geographical entries through the continent-field filter, 288,347 observations covering 243 geographical locations were retained.

B. Selected Variables

The following variables were selected for the main analysis:

- `iso_code`: geographical identifier.
- `location`: country or geographical entity name.
- `date`: observation date.
- `people_fully_vaccinated_per_hundred`: percentage of the population that completed the initial vaccination protocol.
- `new_deaths_smoothed_per_million`: seven-day smoothed new deaths per million people.

Fully vaccinated population percentage was selected to represent vaccination coverage. The smoothed mortality measure was used to reduce the effect of short-term fluctuations in daily reporting.

The smoothed mortality variable was taken from the supplied dataset; it was not recalculated during this analysis.

V. Data Preprocessing

Data preparation was performed before statistical calculations and visualization.

The initial checks included dataset dimensions, column names, data types, date coverage, missing values, duplicate records and geographical identifiers.

A. Data Merging

The vaccination and mortality datasets were combined using the common geographical and temporal identifiers.

Before merging, duplicate country-date keys were examined. No duplicate records were identified in either input dataset.

The inner join retained matching geographical and date combinations.

B. Geographical Filtering

Aggregate entries were excluded using the continent field.

This process reduced the number of geographical locations from 255 to 243 and produced a working dataset containing 288,347 rows.

The term geographical location is used in this study because the supplied dataset contains both countries and other geographical entities.

C. Missing Value Handling

Missing values were inspected in the original datasets.

An unavailable vaccination or mortality measurement was not automatically interpreted as zero.

For each statistical analysis, observations were retained only when the variables required for that specific comparison were available.

No general imputation was applied to the primary statistical series.

D. Coverage Validation

The fully vaccinated population percentage was checked for values exceeding 100 percent.

A total of 228 observations across four locations contained values above this limit:

- Gibraltar: 169 observations.
- Qatar: 37 observations.
- United Arab Emirates: 21 observations.
- Tokelau: 1 observation.

These records were flagged as potential denominator or reporting inconsistencies. They were not silently capped at 100 percent.

A sensitivity analysis was subsequently performed after excluding individual observations exceeding this threshold.

VI. Research Methodology

The analytical workflow consists of dataset preparation, time-series exploration, correlation analysis, time-lagged comparison, weekly-change analysis and interactive visualization.

A. Time-Series Analysis

The vaccination and mortality variables were examined over a common timeline.

For India, fully vaccinated population percentage was compared with seven-day smoothed new deaths per million.

To provide a readable monthly summary, vaccination coverage was represented by the final available observation in each month. Mortality was represented by the monthly mean of the daily smoothed mortality observations.

B. Pearson Correlation

Pearson correlation measures the strength and direction of a linear relationship between two numerical variables.

The coefficient is calculated as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2}$$

where x_i represents vaccination coverage and y_i represents mortality.

The coefficient ranges from -1 to $+1$. A negative value indicates that higher observations of one variable tend to be associated with lower observations of the other variable in the analyzed sample.

C. Spearman Correlation

Spearman correlation examines a monotonic relationship using the ranks of the observations [12].

It is calculated as the Pearson correlation between the ranked variables:

$$\rho = \mathrm{Corr}\left(\mathrm{Rank}(X), \mathrm{Rank}(Y)\right) \tag{3}$$

The use of both Pearson and Spearman correlation provides two complementary descriptions of the observed relationship.

D. Time-Lagged Analysis

Vaccination coverage on one date may be compared with mortality observed after a selected interval.

For a vaccination lead time L, the comparison is represented as:

$$R_L = \mathrm{Corr}(V_{t-L}, M_t) \tag{4}$$

where:

- V_{t-L} denotes vaccination coverage measured L days before the mortality observation.
- M_t represents smoothed new deaths per million on date t.
- L represents the selected vaccination lead interval.

The analysis was conducted using three intervals:

- 14 days.
- 28 days.
- 42 days.

Vaccination and mortality series were matched separately for each interval. This avoids requiring both variables to have valid observations on the same date before lag matching.

E. Weekly-Change Analysis

To examine changes rather than accumulated levels, seven-day differences were calculated for vaccination coverage and mortality.

The weekly change in vaccination coverage is:

$$\Delta_7 V_t = V_t - V_{t-7} \tag{5}$$

The corresponding mortality change is:

$$\Delta_7 M_t = M_t - M_{t-7} \tag{6}$$

The cross-location analysis compared vaccination changes with mortality changes observed 28 days later:

$$R_\Delta = \mathrm{Corr}\left(\Delta_7 V_{t-28}, \Delta_7 M_t\right) \tag{7}$$

The observations were sampled at weekly intervals using Sunday dates.

A minimum of 20 matched observations and variation in both variables were required for a geographical location to be included.

F. Multiple Testing Adjustment

Because statistical tests were conducted separately across several geographical locations, the resulting p-values were adjusted for multiple comparisons.

The Benjamini–Hochberg procedure was used to control the false discovery rate [13].

Pearson and Spearman p-values were adjusted separately. An adjusted p-value below 0.05 was used as the threshold for the exploratory comparison.

This adjustment does not eliminate serial dependence between observations or other sources of statistical bias.

VII. Analytical Workflow

The complete workflow connects the two input datasets with the statistical analysis and the final dashboard.

VIII. Experimental Results and Analysis

The results are organized into three main analytical components:

- India time-series and lagged analysis.
- Cross-location weekly-change analysis.
- Sensitivity analysis of vaccination coverage observations above 100 percent.

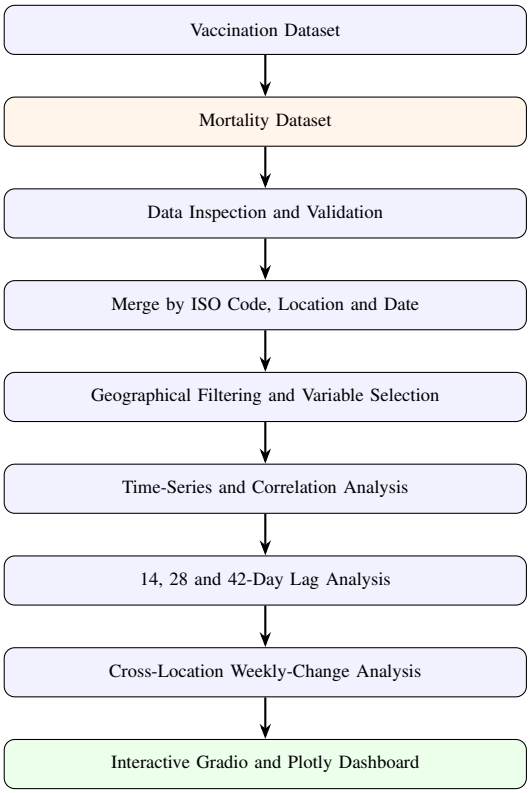


Fig. 1. Workflow used for vaccination and mortality analysis.

A. India Time-Series Analysis

The Indian vaccination series shows a gradual increase in fully vaccinated population coverage beginning in February 2021.

By April 2023, the final available monthly vaccination coverage value was approximately 67.17 percent.

Mortality varied considerably during the same period. The monthly mean of seven-day smoothed deaths per million reached approximately 2.735 in May 2021.

Table II provides selected observations from the monthly series.

TABLE II
SELECTED MONTHLY OBSERVATIONS FOR INDIA

Month	Vaccinated (%)	Deaths per million
Feb. 2021	0.17	0.0717
Apr. 2021	1.88	0.8682
May 2021	3.05	2.7352
Jun. 2021	4.07	1.8078
Dec. 2021	42.57	0.2821
Mar. 2022	58.65	0.1651
Sep. 2022	66.92	0.0202
Apr. 2023	67.17	0.0072

The monthly series shows that mortality was not continuously decreasing as vaccination coverage increased. For example, the major mortality peak occurred during May 2021, followed by substantial changes in later months.

Fig. 2 presents the monthly vaccination and mortality series separately using a shared time reference.

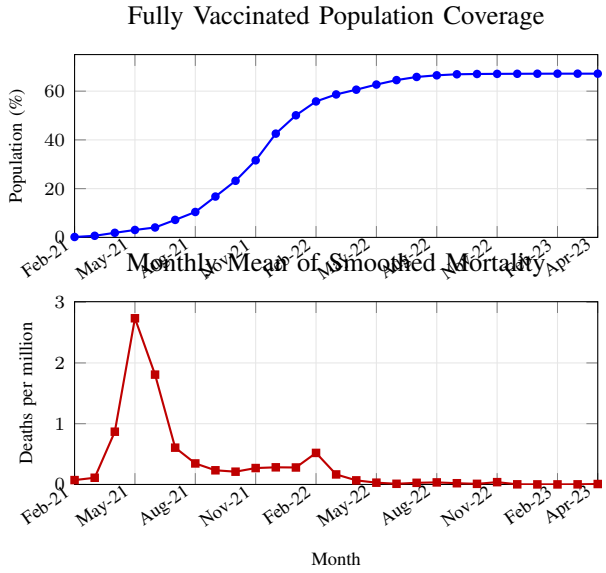


Fig. 2. Monthly vaccination coverage and mortality trends in India from February 2021 to April 2023. Vaccination coverage uses the final available observation in each month, while mortality represents the monthly mean of seven-day smoothed deaths per million.

B. India Time-Lagged Correlation

Pearson and Spearman correlation coefficients were calculated for vaccination lead intervals of 14, 28 and 42 days.

Table III presents the calculated values.

All three lead intervals produced negative correlation coefficients.

Pearson correlation became more negative as the vaccination lead increased from 14 to 42 days. Spearman correlation also indicated a strong negative monotonic association across the selected intervals.

These coefficients describe the relationship between the historical levels of the two variables. They do not demonstrate that vaccination alone caused the observed mortality pattern.

The nominal p-values were very small for all three comparisons. However, the observations are repeated daily time-series measurements and may be serially dependent. Therefore, these nominal p-values should not be treated as independent-observation evidence of causation.

C. Cross-Location Weekly-Change Analysis

The next analysis compared weekly changes in vaccination coverage with weekly changes in mortality 28 days later.

A total of 93 geographical locations met the minimum observation and variation requirements.

Table IV summarizes the direction and median magnitude of the calculated correlations.

The median correlation coefficients were close to zero. Although negative correlations were observed in more than

TABLE III
INDIA: TIME-LAGGED VACCINATION AND MORTALITY CORRELATIONS

Vaccination Lead	Observations	Pearson Correlation	Spearman Correlation	Direction
14 days	751	-0.5871	-0.8718	Negative
28 days	738	-0.6088	-0.8955	Negative
42 days	724	-0.6227	-0.9187	Negative

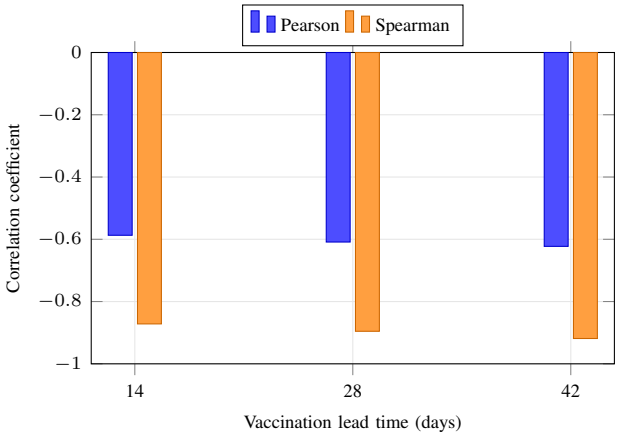


Fig. 3. Pearson and Spearman correlation coefficients for India at the selected vaccination lead intervals.

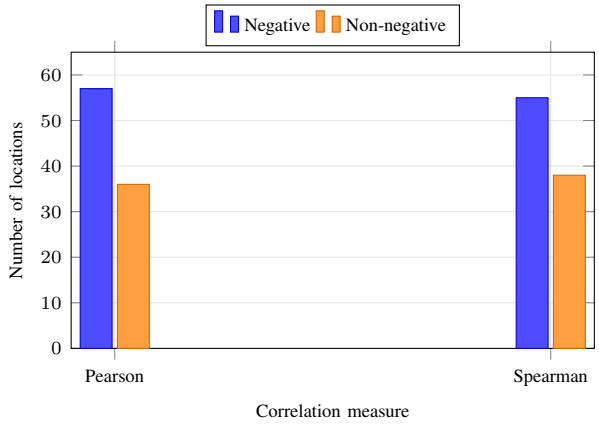


Fig. 4. Distribution of negative and non-negative weekly-change correlations across 93 geographical locations.

TABLE IV
CROSS-LOCATION WEEKLY-CHANGE RESULTS

Measure	Pearson	Spearman
Locations analyzed	93	93
Negative coefficients	57	55
Non-negative coefficients	36	38
Median coefficient	-0.0381	-0.0537
Negative associations passing FDR	3	0

half of the analyzed locations, the overall median magnitudes were small.

After applying the Benjamini–Hochberg adjustment, three negative Pearson associations remained statistically significant at the 0.05 threshold. No negative Spearman association passed the same adjusted threshold.

The three locations with negative Pearson associations passing the adjustment are presented in Table V.

TABLE V
LOCATIONS WITH NEGATIVE PEARSON ASSOCIATIONS PASSING FDR

Location	N	Pearson r	Adjusted q
Colombia	41	-0.6455	0.0005
Kyrgyzstan	62	-0.4826	0.0033
Saudi Arabia	31	-0.6112	0.0081

These results indicate that the strength and direction of the weekly-change association varied between geographical locations.

The country-specific coefficients should not be interpreted as proof that the same relationship applies to other populations.

D. Sensitivity Analysis

The original dataset contained 228 observations with fully vaccinated population percentages exceeding 100 percent.

These observations were flagged rather than automatically capped.

A sensitivity analysis was performed by excluding individual observations above 100 percent and repeating the weekly-change calculations.

After this filtering, 92 geographical locations satisfied the minimum observation requirements.

The median Pearson correlation changed from -0.0381 to -0.0340, while the median Spearman correlation changed from -0.0537 to -0.0536.

The number of negative Pearson associations passing the false discovery rate adjustment remained three. No negative Spearman association passed the adjustment.

TABLE VI
SENSITIVITY ANALYSIS OF COVERAGE VALUES ABOVE 100%

Measure	Original	Filtered
Locations	93	92
Median Pearson	-0.0381	-0.0340
Median Spearman	-0.0537	-0.0536
Negative Pearson passing FDR	3	3
Negative Spearman passing FDR	0	0

The relatively small change in median correlation suggests that excluding the flagged observations did not substantially alter the overall median relationship in this analysis.

However, the presence of values above 100 percent still indicates that population denominators and source reporting require careful validation.

IX. Interactive Dashboard Implementation

The statistical analysis is presented through an interactive dashboard developed using Python, Gradio and Plotly.

The purpose of the dashboard is to allow users to explore vaccination and mortality relationships without directly examining the underlying CSV files.

The implementation contains several interactive components.

A. Country Selection

Users can select a country or geographical location from a dropdown control.

The selected location determines the vaccination and mortality observations used in the displayed charts and calculations.

B. Date Range Selection

Start and end date controls allow users to restrict the analysis to a selected historical period.

The displayed trends and correlation calculations are updated according to the selected period.

C. Vaccination Lead Selection

The dashboard supports three vaccination lead intervals: 14, 28 and 42 days.

Changing the selected interval updates the matched observations and the corresponding correlation results.

D. Summary Indicators

Three KPI cards display:

- Latest available fully vaccinated population percentage within the selected period.
- Latest available smoothed new deaths per million.
- Mean smoothed new deaths per million within the selected period.

E. Interactive Visualizations

The dashboard contains:

- Vaccination and mortality time-series visualization.
- Scatter plot comparing earlier vaccination coverage with subsequent mortality.
- Pearson and Spearman correlation values.
- Comparison of correlation coefficients across the three vaccination lead intervals.

The dashboard therefore combines the data-selection process and statistical output within a single interface.

All results remain descriptive. The dashboard does not estimate a causal vaccination effect or provide individual medical guidance.

X. Discussion

The results show that the observed relationship between vaccination coverage and mortality depends on the analytical method and geographical scope.

For India, the level-based time-lagged analysis produced negative Pearson and Spearman correlations at all three selected vaccination lead intervals.

The magnitude of the negative association was stronger for Spearman correlation than for Pearson correlation.

However, the weekly-change analysis across 93 geographical locations produced median correlation coefficients close to zero.

This difference is an important observation. Vaccination coverage generally increased over time during rollout, whereas mortality varied during different epidemic periods.

Consequently, a strong relationship between the observed levels of two variables may partly reflect their common time progression. The weekly-change analysis offers a different view by examining short-term changes rather than their accumulated values.

The cross-location findings also show that the direction of the relationship was not identical across geographical locations. Differences in population characteristics, reporting practices, vaccination schedules and epidemic conditions may contribute to this variation.

The results must be interpreted alongside existing vaccine effectiveness research [5]–[9]. Those studies employ different designs and measurements, so their estimates cannot be directly compared with the ecological correlations calculated here.

The analysis therefore supports the importance of examining both time-lagged levels and short-term changes when studying public health time series.

XI. Limitations

Several limitations affect the interpretation of this study.

First, the analysis uses aggregated geographical data. It does not contain individual vaccination histories or individual mortality outcomes. Therefore, the observed relationships cannot be used to estimate individual-level vaccine effectiveness.

Second, mortality reporting practices differ between geographical locations. Differences in death registration, testing availability and population estimates can influence the recorded values. Research on excess mortality has demonstrated that reported COVID-19 deaths may not fully represent the pandemic's total mortality impact [10].

Third, fully vaccinated population percentages exceeded 100 percent in a limited number of observations. Although a sensitivity analysis was performed, these inconsistencies indicate that denominator quality requires further investigation.

Fourth, the study does not jointly control for other factors that may influence mortality, including age distribution, previous infections, circulating variants, healthcare availability and public health interventions.

Fifth, repeated observations in daily and weekly time series may be serially dependent. Pearson and Spearman correlation

tests applied to such observations may produce p-values that do not fully account for temporal dependence.

False discovery rate adjustment addresses multiple testing but does not resolve serial dependence. More advanced time-series inference, including autocorrelation-consistent estimation and block-bootstrap procedures, could provide additional uncertainty assessment in future work [14], [15].

Finally, the study uses historical data and should be interpreted as retrospective exploratory analysis rather than real-time monitoring or causal evaluation.

XII. Conclusion and Future Work

This study investigated the relationship between COVID-19 vaccination coverage and mortality using publicly available geographical time-series data.

Vaccination and mortality datasets were combined using ISO geographical identifiers, location names and dates. The resulting analytical workflow included data inspection, geographical filtering, time-series visualization, Pearson and Spearman correlation, time-lagged comparison and cross-location weekly-change analysis.

For India, negative correlations were observed for 14-day, 28-day and 42-day vaccination lead intervals.

However, the cross-location weekly-change analysis showed median coefficients close to zero. Only three negative Pearson associations remained significant after false discovery rate adjustment, while no negative Spearman association passed the same threshold.

The results demonstrate that a relationship observed in the historical levels of vaccination and mortality does not necessarily remain equally strong when their weekly changes are examined.

An interactive dashboard was implemented to support country selection, date filtering, vaccination lead selection and statistical visualization. This allows users to explore the observed patterns across different geographical locations and time periods.

Future work can extend the analysis through demographic stratification, vaccination type, excess mortality indicators and additional public health variables. Statistical approaches that account for serial dependence and potential confounding would also improve the strength of the conclusions.

The present implementation provides an exploratory framework for examining vaccination and mortality relationships while maintaining a distinction between observed statistical association and causal interpretation.

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